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Uncertainty-aware structural trend-decoupling knowledge tracing

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ABSTRACT

Knowledge tracing (KT) infers students' evolving mastery from interaction sequences and serves as a core component of intelligent tutoring systems. Although recent Transformer-based and probabilistic KT models achieve strong predictive performance, they often entangle long-term learning trends with short-term behavioral fluctuations, while inadequately modeling forgetting dynamics under irregular real-time intervals. To address these issues, we propose UST-KT, an uncertainty-aware probabilistic knowledge tracing framework. Specifically, UST-KT introduces a structural cognitive decoupling module that employs causal convolution to separate smooth cognitive trends from high-frequency fluctuations, injecting them into the mean and covariance of Gaussian latent states, respectively. In this way, the mean captures stable long-term mastery, whereas the covariance serves as a proxy for fluctuation-related uncertainty. Furthermore, we develop a Hawkes–Wasserstein attention mechanism. This mechanism modulates Wasserstein-distance-based distributional similarity using a Hawkes-inspired continuous-time decay kernel, enabling more time-aware modeling of forgetting dynamics under irregular time gaps. Extensive experiments on three real-world benchmark datasets demonstrate that UST-KT consistently outperforms strong baselines and yields more stable prediction dynamics, better-calibrated predictive probabilities, together with more structured covariance representations that serve as proxies for fluctuation-related unpredictability.

1. Introduction

Consumer digital platforms such as online learning systems, mobile tutoring applications, and intelligent educational services have become an integral part of everyday user experiences [1]. These systems deliver personalized content and real-time feedback at scale, relying on accurate modeling of user behavior to support adaptive interactions [2]. As a result, learner modeling plays a critical role in enabling reliable and user-centric educational services within modern educational service environments. As a representative approach to learner modeling, Knowledge Tracing (KT) aims to infer students' latent mastery of knowledge concepts (KCs) from their interaction histories with learning systems. As a fundamental component of Intelligent Tutoring Systems (ITS), KT supports personalized instruction, learning diagnosis, and adaptive educational services in modern digital learning platforms [3,4]. Early KT approaches, such as Bayesian Knowledge Tracing (BKT) [4], formulate student learning as a hidden Markov process with explicit state transitions and observation models. With the rapid development of deep learning, KT has evolved toward neural sequence models, including recurrent architectures [5,6] and more recently Transformer-based frameworks. In particular, attention-based KT models [7,8] achieve strong performance by selectively aggregating historical interactions and capturing long-range dependencies in learning trajectories. Beyond sequential models, graph-based approaches have also been applied to educational data mining, demonstrating the effectiveness of relational modeling for learner modeling and knowledge tracing [9]. Complementing these sequential tracking models, alternative paradigms within the broader field of educational measurement, such as the CDMFKC model [10], advance cognitive diagnosis by explicitly focusing on the varying effects of distinct knowledge concepts to profile participant strengths and weaknesses.

Despite their promising results, improving prediction accuracy alone is insufficient for real-world ITS deployments. In practical educational environments, student responses often contain noisy behaviors such as guessing, careless slips, or temporary strategy changes [11,12]. Therefore, KT models should not only provide accurate predictions but

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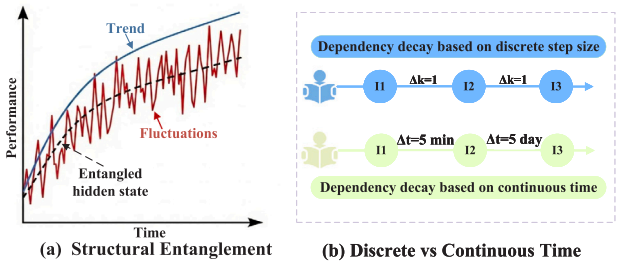


Fig. 1. Illustration of two structural challenges in knowledge tracing. (a) Structural entanglement: student performance sequences contain a smooth long-term learning trend and high-frequency short-term fluctuations, but many KT models entangle them in a single hidden state, making estimates sensitive to behavioral noise. (b) Discrete vs. continuous time: step-based positional encoding induces attention/decay mainly by discrete step distance, and thus cannot distinguish interactions with the same step distance but very different real-time gaps, leading to different forgetting effects that are not properly captured.

also produce reliable and stable estimates of student knowledge states. Such reliability is particularly important in consumer educational applications, where unstable predictions may lead to inappropriate content recommendation or misleading learning feedback. For example, downstream applications such as explainable exercise recommendation [13] and personalized learning path recommendation [14] both rely on reliable knowledge state estimates to make appropriate suggestions; unstable or poorly calibrated predictions would directly degrade the quality of such recommendations. To address this issue, recent probabilistic KT methods represent latent knowledge states as probability distributions and explicitly model uncertainty, which improves robustness and enables uncertainty-aware decision making [15].

However, a closer examination of real-world interaction data reveals two fundamental structural challenges that remain insufficiently addressed in existing deep and probabilistic KT models, as illustrated in Fig. 1. First, student performance sequences typically contain two intertwined components: a smooth long-term cognitive trend reflecting gradual knowledge acquisition, and high-frequency short-term fluctuations caused by guessing, slips, or transient strategy shifts. Most KT models learn hidden representations directly from raw interaction sequences, which inherently conflates gradual cognitive gains with transient behavioral noise. Consequently, short-term noise may be misinterpreted as genuine knowledge changes, while long-term learning progress may be obscured by transient variations, leading to unstable predictions and reduced interpretability [16,17]. Although some recent studies attempt to introduce decomposition-inspired modules [18], such separation is typically performed at the feature level and lacks explicit probabilistic interpretation. As a result, the semantics of distribution parameters remain ambiguous: ideally, the mean should represent expected knowledge mastery, while the covariance captures stochastic variability and serves as a proxy for fluctuation-related uncertainty.

Second, time in student learning processes is inherently irregular [19]. Most Transformer-based KT models rely on positional encoding to represent temporal relationships [8,20], which effectively captures discrete interaction order but fails to distinguish interactions separated by minutes from those separated by days. This limitation restricts the modeling of real-world forgetting dynamics. Continuous-time approaches [21] incorporate temporal decay using point processes, yet they may weaken the inductive bias provided by discrete practice order. Consequently, existing KT models often struggle to jointly exploit discrete sequence structure and continuous-time dynamics within a unified modeling framework.

To address the above challenges, this paper proposes an uncertainty-aware Structural Trend-decoupling Knowledge Tracing framework, termed UST-KT. The proposed model adopts a probabilistic representation of student knowledge states as Gaussian distributions, in which the reliability of covariance-derived uncertainty proxies depends critically on whether the distribution parameters carry semantically meaningful information. To this end, UST-KT introduces structural trend–fluctuation decoupling to ensure that the mean and covariance encode distinct and interpretable signals, and further develops a Hawkes–Wasserstein attention mechanism to ensure that attention aggregation faithfully reflects continuous-time forgetting dynamics rather than distorting the covariance-based uncertainty representation through temporally misaligned dependencies.

The main contributions of this work are summarized as follows:

- We propose UST-KT, a novel uncertainty-aware knowledge tracing framework designed to bridge the gap between stable cognitive evolution and volatile interaction noise within a unified probabilistic modeling framework.
- We design a Structural Cognitive Decoupling Module that treats student interaction sequences as composite signals. By incorporating causal convolutions as a structural inductive bias, this module explicitly decomposes the learning process into low-frequency trends and high-frequency fluctuations. This architectural design assigns complementary roles to the Gaussian mean and covariance, where the mean models long-term mastery and the covariance captures fluctuation-related variability that serves as a proxy for fluctuation-related uncertainty.
- We introduce a hybrid temporal modeling strategy that synergizes discrete practice order with continuous-time forgetting effects. By modulating Wasserstein-distance-based distributional similarity with a Hawkes-style decay kernel, UST-KT achieves a more time-aware characterization of student knowledge state evolution across irregular time intervals.
- Extensive experiments on three real-world benchmark datasets demonstrate that UST-KT consistently outperforms state-of-the-art baselines. Furthermore, ablation studies and qualitative analyses validate the effectiveness of the structural decoupling and continuous-time modulation mechanisms, providing empirical support for using covariance as a tractable proxy for fluctuation-related uncertainty.

2. Related work

KT has evolved from basic sequential modeling to architectures that integrate temporal dynamics and data robustness. However, existing methods often treat temporal irregularity and interaction uncertainty as isolated factors, frequently introducing them as auxiliary features rather than core structural components. This section reviews recent KT advancements in attention mechanisms, time-aware modeling, and uncertainty estimation, and highlights how UST-KT addresses the corresponding limitations through a unified structural decomposition framework.

2.1. Attention-based knowledge tracing

Attention-based architectures have become a dominant paradigm in knowledge tracing due to their ability to model long-range dependencies in student interaction sequences. Representative models, such as SAKT [7], AKT [8], and SAINT [22], leverage self-attention mechanisms to selectively aggregate historical interactions and capture complex learning patterns. Subsequent studies, including SimpleKT [20], demonstrate that simplified attention designs combined with well-structured item representations can achieve competitive performance. Going beyond vanilla attention architectures, recent endeavors enrich item representations to capture implicit cognitive associations. For instance, MAKT [23] deploys a memory-driven module

to mine latent inter-question correlations, whereas FTCKT [24] injects directed transition-graph topologies into sequential semantics via cross-attention mechanisms.

Despite their effectiveness, most attention-based KT models rely on discrete positional encodings to represent temporal dependencies. While such encodings capture interaction order and practice frequency, they implicitly assume uniform spacing between interactions. This assumption limits their ability to model real-world learning processes, where time intervals between interactions are often irregular and forgetting effects are time-dependent.

2.2. Time-aware knowledge tracing

To address the limitation of discrete-time modeling, time-aware KT approaches incorporate temporal information into model design. Existing methods typically introduce elapsed time through feature augmentation, gating mechanisms, or continuous-time processes. For instance, DKT-forgetting [25] and LPKT [26] integrate time-gap features to capture forgetting dynamics within sequential models. Additionally, the LFBKT model [27] introduces a sequential framework that explicitly integrates both learning and forgetting behaviors into sequential state updates to better reflect true knowledge levels. Similarly, LBKT [28] incorporates multi-dimensional learning behaviors, such as answering speed and attempt counts, to characterize the knowledge acquisition process more deeply. DyGKT [29] introduces continuous-time dynamic graph learning to KT for the first time, employing a dual-time embedding and mailbox mechanism to update memories of students and questions as their relationships evolve. HawkesKT [21] adopts a point-process formulation to model temporal influence via continuous-time intensity functions, while SAINT+ [30] augments Transformer-based KT with temporal embeddings. Recent work also focuses on modeling the temporal evolution of knowledge states and learning evidence. Dia4KT [31] proposes diachronic relationship learning with hierarchical difficulty cues and a time-decay bias in attention to refine knowledge assessment across different practice stages. More recent studies further explore spatiotemporal dependencies and time-aware attention mechanisms [32,33]. Recent advancements have explored combining attention-based embeddings directly with explicit cognitive metrics, such as the KVFKT model [34], which integrates the Ebbinghaus forgetting curve based on interaction timestamps to capture interval-driven state attenuation.

Although these methods improve temporal sensitivity, most of them inject time information either at the feature level or as an additive bias to attention scores, rather than coupling it directly with the similarity computation that determines attention weights. Even when such decay is incorporated, it is often parameterized by discrete step distances or relative positions, which can be insufficient for modeling continuous-time forgetting under highly irregular real-time gaps. Moreover, existing methods often struggle to simultaneously preserve the inductive bias of discrete practice order while faithfully characterizing continuous-time forgetting dynamics.

2.3. Robust and uncertainty-aware knowledge tracing

Student interaction data are inherently noisy due to guessing, careless mistakes, and transient behavioral changes, motivating the development of robustness-aware KT models. Prior work improves stability through regularization, representation refinement, or attention reweighting. For example, DKT+ [6] introduces consistency constraints to stabilize predictions, while FlucKT [18] and DenoiseKT [35] mitigate noise through feature decomposition and denoised attention mechanisms. In addition, contrastive learning has been leveraged to improve representation robustness against noise and sparsity. TCL4KT [36] employs a three-view contrastive framework on heterogeneous graphs to learn high-quality question embeddings, and CRANE [37] further mitigates the instability of random perturbations by introducing structure-preserving adaptive contrastive learning and modeling learning efficiency.

Another line of research focuses on modeling uncertainty using probabilistic representations. Methods such as UKT [15] represent knowledge states as Gaussian distributions and leverage distributional similarity for attention computation. Subsequent works [38,39] further enhance robustness through regularization and consistency learning under noisy interactions.

However, existing approaches typically address robustness, decomposition, or uncertainty in isolation. They do not explicitly enforce a structural separation between long-term learning trends and short-term fluctuations within the latent knowledge state. In addition, the integration of uncertainty modeling with time-aware attention mechanisms remains underexplored.

In contrast, this work argues that reliable covariance-derived uncertainty representation rather than probabilistic parameterization alone. Specifically, representing knowledge states as Gaussian distributions is insufficient for meaningful uncertainty modeling unless the distribution parameters are assigned semantically distinct and interpretable roles. To this end, UST-KT enforces a trend–fluctuation separation that gives the mean and covariance complementary physical interpretations, and integrates continuous-time decay directly into distributional similarity to prevent temporally misaligned attention from distorting the covariance-based uncertainty representation.

3. Preliminaries

3.1. Problem definition

Given a student interaction sequence

$$\mathcal{X}_{1:T} = \{(q_k, c_k, r_k, t_k)\}_{k=1}^{T}, \tag{1}$$

where q_k denotes the exercised item, c_k is the associated knowledge concept (KC), $r_k \in \{0, 1\}$ indicates correctness, and t_k is the timestamp.

Following standard KT preprocessing [8,15], each interaction is mapped to a single KC, i.e., $c_k = m(q_k)$. The objective is to predict the probability of a correct response for the next interaction:

$$P(r_{T+1} = 1 \mid q_{T+1}, c_{T+1}, \mathcal{X}_{1:T}). \tag{2}$$

3.2. Gaussian knowledge representation

The latent knowledge state at step k is modeled as a diagonal Gaussian distribution:

$$\mathbf{z}_k \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad \boldsymbol{\Sigma}_k = \mathrm{diag}(\mathbf{s}_k), \tag{3}$$

where $\boldsymbol{\mu}_k \in \mathbb{R}^d$ represents the mean and $\mathbf{s}_k \in \mathbb{R}_+^d$ denotes the variance.

To ensure numerical stability, $\mathbf{s}_k$ is constrained to be positive using $\mathrm{ELU}(\cdot) + 1$ followed by a small-value clamp.

3.3. Wasserstein distance

For two diagonal Gaussian states $\mathbf{z}_i$ and $\mathbf{z}_j$, the squared 2-Wasserstein distance is computed as:

$$W_2^2(\mathbf{z}_i, \mathbf{z}_j) = \|\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\|_2^2 + \|\sqrt{\mathbf{s}_i} - \sqrt{\mathbf{s}_j}\|_2^2. \tag{4}$$

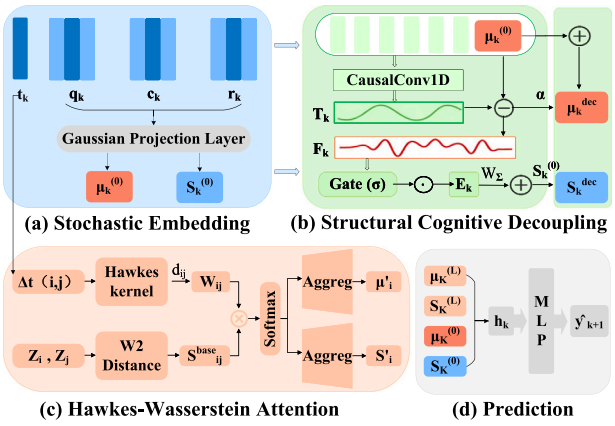


Fig. 2. Overall architecture of UST-KT: stochastic Gaussian embedding, structural cognitive decoupling, and Hawkes–Wasserstein attention for continuous-time modulated temporal aggregation.

4. Proposed method

UST-KT is an end-to-end probabilistic knowledge tracing framework that improves student learning modeling by explicitly separating long-term cognitive trends from short-term behavioral fluctuations, while jointly modeling discrete practice order and continuous-time gaps to capture forgetting effects. As illustrated in Fig. 2, UST-KT first encodes each interaction into a stochastic latent state represented by a Gaussian distribution, capturing both expected mastery and latent variability. These probabilistic states are then processed by a structural cognitive decoupling module that extracts smooth trends and residual fluctuations, injecting them into the Gaussian mean and covariance respectively to encourage complementary semantics. After decoupling, fixed cosine positional encodings are added to preserve the inductive bias of discrete practice order. Subsequently, we perform temporal context aggregation via a Hawkes–Wasserstein attention mechanism, which integrates Wasserstein-distance-based distributional similarity with a Hawkes-style continuous-time modulation to account for irregular intervals and time-decaying effects. Ultimately, the refined knowledge representation is utilized by the prediction layer to estimate the student's performance on the upcoming task.

4.1. Stochastic interaction embedding

Following prior probabilistic KT models [15], we adopt a standard stochastic embedding layer to encode each interaction as a diagonal covariance Gaussian state. Specifically, we learn separate embeddings for the mean and variance. Let $\mathbf{e}^{\mu}(c_k), \mathbf{e}^{s}(c_k) \in \mathbf{R}^d$ denote the KC mean/variance embeddings, and $\mathbf{e}^{\mu}(r_k), \mathbf{e}^{s}(r_k) \in \mathbf{R}^d$ denote the response mean/variance embeddings. We initialize the interaction state by

$$\boldsymbol{\mu}_k^{(0)} = \mathbf{e}^{\mu}(c_k) + \mathbf{e}^{\mu}(r_k), \quad \mathbf{s}_k^{(0)} = \mathbf{e}^{s}(c_k) + \mathbf{e}^{s}(r_k), \tag{5}$$

yielding $\mathbf{z}_k^{(0)} \sim \mathcal{N}(\boldsymbol{\mu}_k^{(0)}, \mathrm{diag}(\phi(\mathbf{s}_k^{(0)})))$, where $\phi(\cdot)$ enforces positivity (e.g., ELU+1 with a small clamp).

When problem IDs q_k are available, we optionally apply Rasch-style difficulty modulation [8] to refine representations; the remaining components of UST-KT are built on top of these stochastic embeddings. Positional information is intentionally withheld at this stage, ensuring that the subsequent decoupling module operates on behavioral signals rather than a positional ordering prior.

4.2. Structural cognitive decoupling

Given the mean sequence $\{\boldsymbol{\mu}_k^{(0)}\}_{k=1}^T$ and variance sequence $\{\mathbf{s}_k^{(0)}\}_{k=1}^T$, UST-KT introduces a structural inductive bias that explicitly decouples long-term cognitive trends from short-term behavioral fluctuations. Unlike conventional probabilistic models that rely on implicit mapping to generate distribution parameters, our module enforces this separation at the architectural level by treating the input as a composite signal.

Specifically, we view $\{\boldsymbol{\mu}_k^{(0)}\}$ as a noisy cognitive signal and extract its low-frequency component using a causal 1D convolution followed by layer normalization:

$$\mathbf{T}_k = \mathrm{LN}\big(\mathrm{CausalConv1D}(\boldsymbol{\mu}^{(0)})_k\big), \tag{6}$$

where causal padding ensures $\mathbf{T}_k$ depends only on past and current steps $\{\boldsymbol{\mu}_j^{(0)}\}_{j\le k}$. We then define the residual fluctuation as

$$\mathbf{F}_k = \boldsymbol{\mu}_k^{(0)} - \mathbf{T}_k. \tag{7}$$

To suppress irrelevant noise, we apply an element-wise gate to the residual:

$$\mathbf{g}_k = \sigma(\mathbf{W}\mathbf{F}_k + \mathbf{b}), \tag{8}$$

and compute the non-negative fluctuation energy

$$\mathbf{E}_k = (\mathbf{F}_k \odot \mathbf{g}_k)^{\odot 2}. \tag{9}$$

We update the Gaussian mean via bounded residual fusion, which encourages temporal smoothness while preserving individual-specific information:

$$\boldsymbol{\mu}_k^{\mathrm{dec}} = \boldsymbol{\mu}_k^{(0)} + \alpha_{\mu}(\mathbf{T}_k - \boldsymbol{\mu}_k^{(0)}), \tag{10}$$

where $\alpha_{\mu} = \sigma(\theta_{\mu}) \cdot \tau_{\mu}$ is learnable and τ_{μ} is an upper bound (default $\tau_{\mu} = 0.5$). In parallel, we inject fluctuation energy into the variance through a bounded residual update:

$$\mathbf{s}_k^{\mathrm{dec}} = \phi(\mathbf{s}_k^{(0)}) + \beta_{\Sigma} \cdot \tanh(\mathbf{W}_{\Sigma}\mathbf{E}_k) \cdot \tau_{\Sigma}, \tag{11}$$

where $\beta_{\Sigma} = \sigma(\theta_{\Sigma})$ is learnable, and τ_{Σ} controls the maximal update magnitude (default $\tau_{\Sigma} = 0.5$). The resulting variance is clamped to a small positive constant for numerical stability.

By assigning the smoothed trend $\mathbf{T}_k$ to the mean and the gated fluctuation energy $\mathbf{E}_k$ to the variance, we ensure that the Gaussian parameters possess distinct physical semantics, where the mean captures stable knowledge mastery and the variance serves as a structural proxy for transient behavioral uncertainty. This design also helps prevent degenerate or exploding variances, which is critical for stable Wasserstein distance computation and reliable attention propagation. After decoupling, fixed cosine positional encodings are added to both the mean and variance sequences, preserving the inductive bias of discrete practice order prior to attention aggregation.

4.3. Hawkes–Wasserstein attention

UST-KT aggregates historical probabilistic states using an attention mechanism that combines Wasserstein-distance-based distributional similarity with continuous-time decay for modeling irregular time gaps and forgetting effects. Instead of formulating student interactions as a full Hawkes point process [40], we only adopt a Hawkes-inspired temporal decay form to modulate attention scores according to elapsed time.

For diagonal-covariance Gaussian states, we first compute the base similarity score between step i and a historical step j as:

$$S_{ij}^{\mathrm{base}} = -\frac{1}{\sqrt{d_k}} W_2^2(\mathbf{z}_i, \mathbf{z}_j), \tag{12}$$

where d_k is the per-head dimension and $\mathbf{z}_i = \mathcal{N}(\boldsymbol{\mu}_i, \mathrm{diag}(\mathbf{s}_i))$.

Compared with dot-product similarity on deterministic embeddings, the Wasserstein distance jointly measures discrepancies in both mean

and uncertainty, enabling attention to capture distributional differences between latent cognitive states.

The similarity score is further modulated by a continuous-time decay factor so that interactions separated by larger elapsed time receive weaker influence. For $i \geq j$, let $\Delta t_{ij} = \max(t_i - t_j, 0)$ be the elapsed time gap, and we apply log-compression with a time-scale hyperparameter $\gamma > 0$:

$$\tilde{\Delta} t_{ij} = \log\left(1 + \frac{\Delta t_{ij}}{\gamma}\right). \tag{13}$$

We define a decay term

$$d_{ij} = \frac{1}{1 + \rho\, \tilde{\Delta} t_{ij}}, \quad \rho = \mathrm{softplus}(\beta_t) + \epsilon_\beta, \tag{14}$$

where $\epsilon_\beta \geq 0$ avoids degenerate decay rates, and obtain the time modulation weight

$$w_{ij} = 1 - \sigma(\alpha_t)(1 - d_{ij}). \tag{15}$$

Finally, the decay weight is applied multiplicatively to the pre-softmax attention score, allowing temporally distant interactions to contribute less during attention aggregation:

$$S_{ij}^{\text{final}} = S_{ij}^{\text{base}} \cdot w_{ij}. \tag{16}$$

Note that since $S_{ij}^{\text{base}} \leq 0$, scaling it by $w_{ij} \in (1 - \sigma(\alpha_t),\, 1]$ shrinks its absolute magnitude, i.e., the score is pulled toward zero. This induces an asymmetric effect across time: temporally close pairs ($w_{ij} \approx 1$) largely preserve the original score range, whereas temporally distant pairs ($w_{ij} \ll 1$) are strongly compressed toward zero, thereby collapsing their inter-score variance. Under softmax normalization, this reduced separability makes distant interactions contribute a nearly uniform and collectively low-weight component, while recent interactions retain sharper score differentiation and thus attract more attention mass. The learnable gate $\sigma(\alpha_t)$ adaptively controls the decay strength: as α_t decreases, $w_{ij} \to 1$ and the temporal modulation is effectively disabled; as α_t increases, distant interactions are attenuated more aggressively.

With a causal mask, each position attends only to past interactions, and attention weights are computed as:

$$a_{ij} = \frac{\exp(S_{ij}^{\text{final}})}{\sum_{j' \leq i} \exp(S_{ij'}^{\text{final}})}. \tag{17}$$

Under the diagonal covariance assumption, we propagate the first and second moments by:

$$\begin{aligned} \boldsymbol{\mu}_i' &= \sum_{j \leq i} a_{ij} \boldsymbol{\mu}_j, \\ \mathbf{s}_i' &= \sum_{j \leq i} a_{ij}^2\, \mathbf{s}_j, \end{aligned} \tag{18}$$

which corresponds to an efficient approximation of moment propagation under the diagonal covariance assumption, where latent dimensions are treated independently.

By stacking L attention layers, UST-KT obtains the final context-aware Gaussian states.

4.4. Prediction and optimization

Let $(\boldsymbol{\mu}_k^{(L)}, \mathbf{s}_k^{(L)})$ denote the Gaussian parameters produced by the L-layer Hawkes–Wasserstein attention stack at position k. Following the implementation, we concatenate the context-aware distribution parameters with the corresponding input stochastic embedding. Let the combined representation be:

$$\mathbf{h}_k = [\boldsymbol{\mu}_k^{(L)}; \mathbf{s}_k^{(L)}; \boldsymbol{\mu}_k^{(0)}; \mathbf{s}_k^{(0)}]. \tag{19}$$

We then feed it into an MLP to predict the probability of a correct response at the next step:

$$\hat{y}_{k+1} = \sigma\Big(\mathrm{MLP}(\mathbf{h}_k)\Big), \tag{20}$$

Table 1
Statistics of the evaluated datasets.

Dataset	#Stu.	#Ques.	#KCs	#Ints.	KC/Q
BD2006	9680	129 263	493	1 824 310	1.0136
NIPS34	9401	948	57	1 399 470	1.0137
AL2005	4712	173 113	112	884 098	1.3521

where $\sigma(\cdot)$ is the sigmoid function.

The model is trained end-to-end using binary cross-entropy loss, computed only over valid (non-padding) positions. Let $m_k \in 0, 1$ be a binary mask indicating whether the prediction target at step k+1 is valid. The optimization objective is:

$$\begin{aligned} \mathcal{L} = -\frac{1}{\sum_{k=1}^{T-1} m_k} \sum_{k=1}^{T-1} m_k \Big[& r_{k+1} \log \hat{y}_{k+1} \\ & + (1 - r_{k+1}) \log(1 - \hat{y}_{k+1}) \Big]. \end{aligned} \tag{21}$$

5. Experiments

The experiments are designed to investigate the following research questions (RQs):

- **RQ1:** How does UST-KT perform overall compared with baseline methods across multiple benchmark datasets?
- **RQ2:** How do the variations in convolution kernel size and the time-scale hyperparameter γ affect the predictive performance of UST-KT across diverse datasets?
- **RQ3:** Does UST-KT exhibit more stable prediction dynamics, and can the learned covariance be interpreted as a meaningful carrier of fluctuation-related uncertainty?
- **RQ4:** Are the key components in UST-KT effective?

5.1. Datasets

UST-KT is evaluated on three public knowledge tracing benchmarks with timestamps. Detailed statistics, including the number of students (Stu.), questions (Ques.), knowledge concepts (KCs), interactions (Ints.), and the average number of knowledge concepts per question (KC/Q), are summarized in Table 1.

1. Bridge to Algebra 2006 (BD2006): Derived from the KDD Cup 2010 EDM Challenge [41], this dataset contains student interactions collected from an intelligent tutoring system.
2. NeurIPS 2020 Education Challenge (NIPS34): Released by the NeurIPS 2020 Education Challenge [42], this dataset contains student responses to mathematics problems from the Eedi platform. We use the subsets corresponding to Tasks 3 and 4.
3. Algebra 2005 (AL2005): Also from the KDD Cup 2010 EDM Challenge [41], this dataset provides fine-grained, step-level student responses to algebra problems.

Following the evaluation protocol in PyKT [43], each dataset is preprocessed with student-level splitting to avoid information leakage across users, and we remove student sequences containing fewer than three interactions.

5.2. Baselines

UST-KT is compared with representative KT baselines covering major modeling paradigms, including recurrent KT, attention-based KT, continuous-time KT, and probabilistic KT:

- **DKT** [5] employs recurrent neural networks to model student learning trajectories and updates latent knowledge states sequentially from historical interactions.

- **DKT-Forget** [25] extends DKT by incorporating temporal interval and practice-frequency features to model forgetting-related effects in student learning sequences.
- **LPKT** [26] models learning gains and forgetting dynamics separately through dedicated modules, enabling process-consistent knowledge state evolution.
- **SAKT** [7] introduces self-attention into knowledge tracing and selectively aggregates relevant historical interactions through scaled dot-product attention.
- **AKT** [8] adopts a Transformer-based architecture with contextual and monotonic attention mechanisms to model exercise relations and temporal dependencies.
- **SAINT** [22] employs an encoder–decoder Transformer framework to jointly model exercise sequences and response sequences for large-scale knowledge tracing.
- **SimpleKT** [20] simplifies Transformer-based KT using lightweight attention and Rasch-style interaction representations while maintaining competitive performance.
- **HawkesKT** [21] introduces Hawkes processes into KT to model continuous-time learning dynamics and temporal forgetting effects.
- **UKT** [15] represents latent knowledge states as Gaussian distributions and employs Wasserstein-distance-based attention for uncertainty-aware knowledge tracing.
- **FlucKT** [18] separates long-term cognitive trends and short-term fluctuations through causal convolutional decomposition to improve robust sequence modeling.

Whenever possible, the authors' publicly available implementations are used; otherwise, the baselines are reimplemented following the PyKT protocol [43]. All models are tuned on the validation set under identical data splits and search spaces. Weights & Biases is used to perform Bayesian hyperparameter sweeps and log all runs. Since the official implementation of FlucKT is not publicly available, its results are reported from the original paper for reference.

5.3. Implementation details

The standard experimental protocol in PyKT [43] is followed with student-level partitioning. Specifically, 20% of students are held out as the test set, and the remaining 80% are used for 5-fold cross-validation for model selection. All sequences are truncated or padded to a maximum length of 200. Models are trained for up to 200 epochs using Adam, with early stopping (patience 10) on the validation split.

We perform Bayesian hyperparameter sweeps using Weights & Biases over key hyperparameters within the following ranges: embedding/hidden size $\in \{64, 128, 256, 512\}$, number of Transformer layers $L \in \{2, 4\}$, number of heads $\in \{4, 8\}$, learning rate $\in \{10^{-3}, 10^{-4}, 10^{-5}\}$, and dropout $\in \{0.05, 0.1, 0.3, 0.5\}$. For the time-scale parameter γ in continuous-time modulation, we select the best value from a predefined candidate set, and RQ2 further studies sensitivity to γ. Each sweep evaluates a large number of configurations per model and dataset, and the best configuration is selected based on validation performance. All methods are implemented in PyTorch, and the entire experimental pipeline can be completed within 40 min on a single NVIDIA V100 GPU.AUC and ACC are reported as mean±std over multiple random seeds.

5.4. Overall performance (RQ1)

Tables 2 and 3 report the overall performance of all models on three benchmark datasets. We report the mean and standard deviation of AUC and ACC over multiple runs under the same evaluation protocol. Based on Tables 2–3, we have the following observations.

On the primary metric AUC, UST-KT achieves 0.8290, 0.8058, and 0.9337 on BD2006, NIPS34, and AL2005, respectively; compared with the second-best model on each dataset, it improves AUC by 0.21, 0.11, and 0.28 percentage points. On the secondary metric ACC, UST-KT attains the best results on NIPS34 and AL2005, and remains highly competitive on BD2006. We attribute these gains to two factors. First, structural trend–fluctuation decoupling separates smooth long-term cognitive changes from high-frequency noise, stabilizing Gaussian mean estimation while allocating fluctuation-related uncertainty to the covariance. Second, Hawkes–Wasserstein attention injects irregular time gaps into the attention similarity computation, enabling more time-aware modeling of forgetting and thereby improving ranking performance. To further verify that the observed improvements are statistically reliable, we conduct paired statistical significance tests between UST-KT and the strongest baseline UKT across multiple random seeds. The results are reported in Table 4.

Table 2
Overall AUC on three benchmark datasets.

Model	BD2006	NIPS34	AL2005
DKT	0.7981 ± 0.0008	0.7707 ± 0.0002	0.9188 ± 0.0011
DKT-F	0.7975 ± 0.0011	0.7730 ± 0.0003	0.9183 ± 0.0013
LPKT	0.8013 ± 0.0007	0.7903 ± 0.0003	0.9238 ± 0.0004
SAKT	0.7619 ± 0.0008	0.7476 ± 0.0005	0.9052 ± 0.0015
AKT	0.8139 ± 0.0007	0.7980 ± 0.0004	0.9203 ± 0.0019
SAINT	0.7195 ± 0.0014	0.7852 ± 0.0007	0.8750 ± 0.0016
SimpleKT	0.8113 ± 0.0006	0.8032 ± 0.0000	0.9234 ± 0.0003
HawkesKT	0.8024 ± 0.0007	0.5743 ± 0.0006	0.7331 ± 0.0015
UKT	0.8168 ± 0.0010	0.8009 ± 0.0004	0.9309 ± 0.0012
FlucKT	0.8269 ± 0.0005	0.8047 ± 0.0003	0.8376 ± 0.0011
UST-KT	**0.8290** ± 0.0003	**0.8058** ± 0.0003	**0.9337** ± 0.0013

Table 3
Overall ACC on three benchmark datasets.

Model	BD2006	NIPS34	AL2005
DKT	0.8544 ± 0.0008	0.7036 ± 0.0005	0.8675 ± 0.0011
DKT-F	0.8536 ± 0.0004	0.7051 ± 0.0003	0.8660 ± 0.0005
LPKT	0.8547 ± 0.0005	0.7203 ± 0.0003	0.8725 ± 0.0008
SAKT	0.8469 ± 0.0006	0.6841 ± 0.0007	0.8585 ± 0.0020
AKT	0.8546 ± 0.0005	0.7320 ± 0.0005	0.8686 ± 0.0010
SAINT	0.8395 ± 0.0005	0.7164 ± 0.0004	0.8444 ± 0.0005
SimpleKT	0.8551 ± 0.0004	0.7312 ± 0.0002	0.8698 ± 0.0003
HawkesKT	0.8539 ± 0.0008	0.6057 ± 0.0009	0.7854 ± 0.0007
UKT	0.8563 ± 0.0005	0.7290 ± 0.0003	0.8775 ± 0.0005
FlucKT	**0.8614** ± 0.0003	0.7337 ± 0.0004	0.8153 ± 0.0006
UST-KT	0.8612 ± 0.0002	**0.7354** ± 0.0003	**0.8790** ± 0.0005

Table 4
Statistical significance analysis based on paired t-tests between UST-KT and UKT.

Dataset	Metric	UKT	UST-KT	p-value
BD2006	AUC	0.8168 ± 0.0010	**0.8290** ± 0.0003	< 0.001
	ACC	0.8563 ± 0.0005	**0.8612** ± 0.0002	< 0.001
NIPS34	AUC	0.8009 ± 0.0003	**0.8058** ± 0.0002	< 0.001
	ACC	0.7290 ± 0.0003	**0.7354** ± 0.0003	< 0.001
AL2005	AUC	0.9309 ± 0.0011	**0.9337** ± 0.0011	0.0021
	ACC	0.8775 ± 0.0005	**0.8790** ± 0.0005	< 0.001

5.5. Effect of γ and convolution kernel size (RQ2)

UST-KT incorporates temporal information through two complementary mechanisms. First, in the Hawkes–Wasserstein attention, continuous-time gaps are log-compressed using a time-scale hyperparameter γ, which determines how sensitive the attention modulation is to elapsed time (smaller γ implies stronger sensitivity, while larger γ weakens the effect by compressing time-gap differences). Second, in the structural cognitive decoupling module, a causal 1D convolution is applied to the Gaussian mean sequence to extract a smooth long-term trend component; therefore, the convolution kernel size k directly

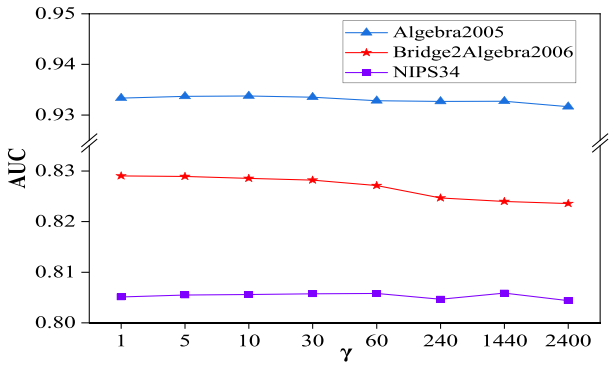


Fig. 3. Sensitivity to the time-scale hyperparameter γ in the log-compressed time-gap encoding. Annotated values indicate the best AUC achieved on each dataset.

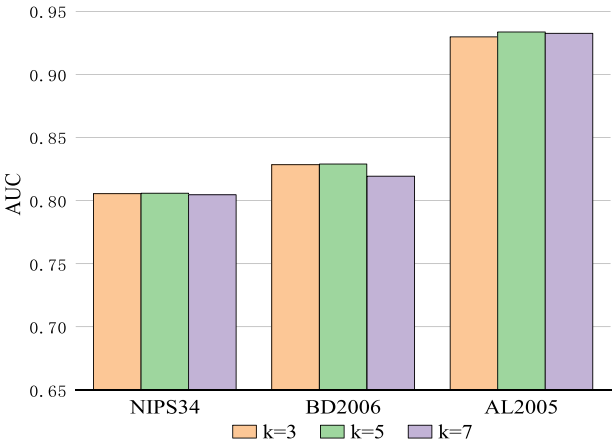


Fig. 4. AUC results across datasets with different causal convolution kernel sizes k for trend extraction.

controls the smoothing range of the extracted trend (smaller k preserves more short-term variation, whereas larger k produces a smoother trend but may over-smooth informative fluctuations).

To answer RQ2, we vary the time-scale hyperparameter $\gamma \in \{1, 5, 10, 30, 60, 240, 1440, 2400\}$ while keeping all other settings fixed, and report the resulting test AUC on BD2006, NIPS34, and AL2005, as shown in Fig. 3. In addition, we investigate the effect of the causal convolution kernel size by evaluating $k \in \{3, 5, 7\}$ in the trend extraction module, with the corresponding AUC results presented in Fig. 4.

Fig. 3 shows that varying γ yields consistent yet dataset-dependent changes. On BD2006, AUC gradually decreases as γ increases (from 0.8290 at $\gamma = 1$ to 0.8235 at $\gamma = 2400$), indicating that stronger time-gap sensitivity is beneficial for long and noisy trajectories where elapsed time likely contains informative forgetting-related patterns. On AL2005, a similar but milder pattern is observed: the best AUC is achieved at $\gamma = 10$ (0.9337) and slightly decreases as γ grows, suggesting that continuous-time modulation remains useful even under denser practice patterns. In contrast, NIPS34 exhibits only minor variation across γ, with the best result obtained at a larger time scale (0.8058 at $\gamma = 1440$), implying that smoother time decay can be preferable under highly irregular time gaps.

From a modeling perspective, γ controls how aggressively real-time gaps are distinguished after log-compression: a smaller γ preserves larger relative differences between short and long gaps, leading to stronger time-aware attenuation; a larger γ smooths these differences and yields a more conservative decay. The dataset-dependent optimal γ in Fig. 3 suggests that the usefulness of fine-grained time-gap discrimination varies across benchmarks. When real-time gaps carry more

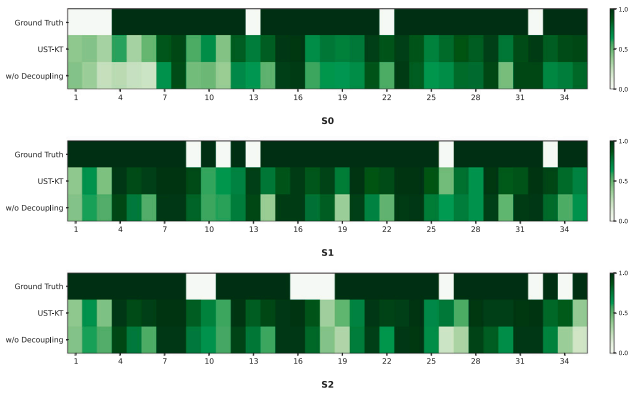


Fig. 5. Predicted correctness probabilities across interaction steps for three students (S0–S2) from the Bridge to Algebra 2006 dataset. The horizontal axis denotes interaction steps, and the vertical axis denotes predicted correctness probability.

consistent forgetting-related signals, stronger sensitivity can be advantageous; when time gaps are extremely irregular or noisy proxies of forgetting, overly aggressive attenuation may suppress useful historical evidence, and a smoother time scale can be more robust.

It is also worth noting that the overall AUC variation across a wide range of γ values remains limited on some datasets, indicating that UST-KT is not overly sensitive to this hyperparameter and maintains robust performance under different time-scaling choices.

In practice, we recommend selecting γ via validation within a coarse candidate set (as used in our implementation) rather than fine-grained tuning, because Fig. 3 shows broad plateaus where multiple γ values achieve comparable AUC. This makes the temporal modulation easy to deploy while still benefiting from continuous-time information when it is predictive.

Fig. 4 examines the kernel size k in the causal convolution trend extractor, which determines the smoothing horizon of the low-frequency component. A small kernel may retain more high-frequency variation in the extracted trend, making the trend channel less distinguishable from short-term behavioral noise. Conversely, an overly large kernel can over-smooth the sequence and attenuate informative mid-term changes (e.g., phase transitions in learning), weakening the intended separation between stable mastery evolution and transient fluctuations. Across the three datasets, a moderate kernel size ($k = 5$) provides the best overall trade-off, while the performance differences remain relatively small, suggesting that the decoupling mechanism is reasonably robust to the exact smoothing horizon as long as it stays within a moderate range.

These results demonstrate that UST-KT is influenced by both the time scale γ and the convolution kernel size, but in different ways: γ controls the sensitivity to irregular elapsed time in attention, whereas the kernel size governs the smoothing scale for trend–fluctuation decoupling. Overall, the best performance is achieved with a moderate kernel size and a dataset-appropriate time scale. These results confirm that γ and the kernel size control distinct aspects of temporal and structural modeling in UST-KT, and that their joint optimization is important for achieving robust performance across datasets with diverse temporal characteristics.

5.6. Qualitative and quantitative analysis (RQ3)

All visualizations are generated with a fixed random seed for reproducibility. We randomly sample three student sequences from the BD2006 test split without manual cherry-picking.

Fig. 5 shows that UST-KT produces prediction trajectories with stronger temporal continuity than the variant *w/o Decoupling* for the shown students (S0–S2), exhibiting fewer abrupt oscillations between

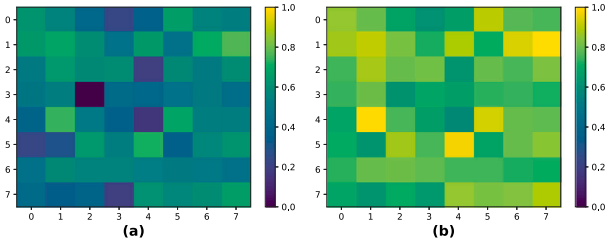


Fig. 6. Normalized variance over a batch of student sequences before and after structural decoupling. Subfigure (a) shows the variance without decoupling, while (b) corresponds to the decoupled setting.

consecutive steps. This behavior is consistent with our structural cognitive decoupling design: causal convolution acts as a learnable low-pass filter on the mean channel, preserving long-term mastery evolution while suppressing high-frequency behavioral noise. In contrast, removing decoupling yields more locally volatile predictions, suggesting that short-term fluctuations are more likely to be entangled with knowledge signals and amplified by downstream attention aggregation.

To qualitatively examine whether the covariance captures fluctuation-related dynamics, Fig. 6 visualizes the normalized diagonal variance before and after decoupling, shown in subfigures (a) and (b), respectively. Without decoupling (a), the variance across latent dimensions shows weak and unstructured temporal patterns. After decoupling (b), the variance exhibits clearer temporal patterns and stronger localized activations across latent dimensions, aligning with our design that allocates short-term residual fluctuations to the covariance channel. This separation allows UST-KT to maintain stable mean estimates while expressing short-term unpredictability through increased variance, rather than requiring the mean representation to absorb transient behavioral fluctuations.

To further validate that the causal convolution produces a structurally meaningful separation between the two components, we conduct a signal-level analysis of the extracted trend and fluctuation sequences across all three datasets. As illustrated in Fig. 7, we measure three complementary properties for each component: *smoothness* (mean absolute difference between consecutive steps), *variance* (overall magnitude of step-to-step variation), and *temporal stability* (autocorrelation function over lags 1–5).

Across all three datasets, the trend component consistently exhibits markedly lower smoothness, lower variance, and higher autocorrelation than the fluctuation component. On BD2006, the smoothness of the trend is 0.135 versus 0.656 for the fluctuation, and the variance is 0.049 versus 0.879. Similar patterns hold on AL2005 (smoothness: 0.209 vs. 0.904; variance: 0.136 vs. 0.955) and NIPS34 (smoothness: 0.120 vs. 0.345; variance: 0.069 vs. 0.934). The autocorrelation curves further confirm that the trend maintains stronger temporal coherence across lags, while the fluctuation decays more rapidly, indicating weaker serial dependency.

These results provide direct signal-level evidence that the causal convolution in UST-KT functions as an effective structural inductive bias: it reliably separates a temporally stable, low-variance component (the trend) from a high-variance, rapidly varying component (the fluctuation). This property supports the design rationale of injecting the trend into the Gaussian mean and the fluctuation energy into the covariance, ensuring that the two distribution parameters carry semantically distinct and empirically grounded representations rather than serving as arbitrary latent channels.

Together, Figs. 5, 6, and 7 provide qualitative support for RQ3: UST-KT yields more stable prediction dynamics, and the decoupled covariance becomes more structured and more aligned with fluctuation-related dynamics.

To examine whether the covariance-derived uncertainty proxy is aligned with empirical prediction errors, we further compute the

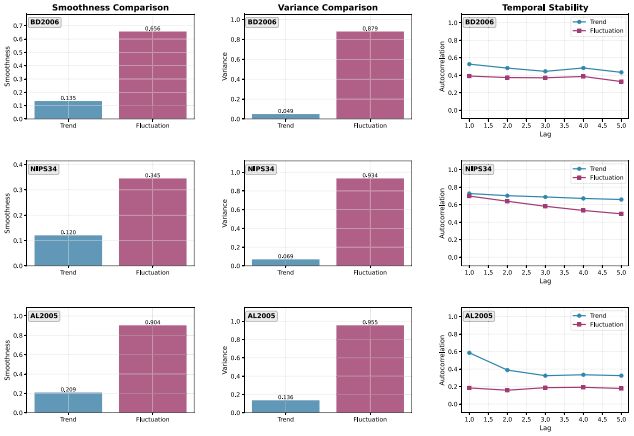


Fig. 7. Signal-level analysis of decoupled trend and fluctuation components across three datasets. The trend consistently exhibits lower smoothness, lower variance, and higher autocorrelation than the fluctuation, confirming a structurally meaningful separation between long-term cognitive stability and short-term behavioral variability.

uncertainty–error correlation, denoted as $\mathrm{Corr}(U, E)$. The correlation is computed at the interaction level after concatenating all valid interactions across the entire test set. For each valid interaction i, the uncertainty proxy U_i is computed as the average predicted diagonal variance extracted from the covariance output of UST-KT after masking padded positions, i.e.,

$$U_i = \frac{1}{d}\sum_{j=1}^{d} \Sigma_{ijj}, \tag{22}$$

where Σ_{ijj} denotes the jth diagonal element of the predicted covariance matrix for interaction i, and d is the latent representation dimension. The empirical prediction error is defined as

$$E_i = \left| y_i - \hat{y}_i \right|, \tag{23}$$

where $y_i \in \{0, 1\}$ denotes the ground-truth response and $\hat{y}_i \in [0, 1]$ is the predicted correctness probability. The overall correlation is computed using Pearson's correlation coefficient:

$$\mathrm{Corr}(U, E) = \frac{\sum_{i=1}^{M}(U_i - \bar{U})(E_i - \bar{E})}{\sqrt{\sum_{i=1}^{M}(U_i - \bar{U})^2 \sum_{i=1}^{M}(E_i - \bar{E})^2}}, \tag{24}$$

where $\bar{U}$ and $\bar{E}$ denote the sample means of U and E, respectively, and M is the total number of valid interaction instances after masking padded positions. A higher positive correlation indicates that the covariance-derived proxy is more strongly aligned with empirical prediction errors, suggesting that it provides a useful proxy for fluctuation-related unpredictability.

Beyond predictive accuracy, Table 5 reports calibration-oriented evaluation of the predictive probabilities produced by UST-KT, together with the uncertainty–error correlation $\mathrm{Corr}(U, E)$ derived from the covariance-based uncertainty proxy. Across all three datasets, UST-KT achieves lower ECE/Brier/NLL and higher $\mathrm{Corr}(U, E)$ than both the strongest probabilistic baseline UKT and all ablated variants, indicating that the proposed structural grounding contributes not only to predictive accuracy but also to better-calibrated predictive probabilities and a covariance-derived uncertainty proxy that is more closely aligned with empirical prediction errors.

The results further reveal how individual components shape calibration quality. Removing structural decoupling (*w/o Decoupling*) leads to consistent degradation in ECE and Corr(U,E) across datasets, suggesting that explicitly separating trends from fluctuations benefits the reliability of probability estimates beyond what AUC alone captures.

Table 5
Calibration performance and uncertainty-error correlation across datasets. Values are reported as mean ± std over 5-fold cross validation. ECE, Brier Score, and NLL are lower-is-better, while Corr(U,E) is higher-is-better.

Model	BD2006				NIPS34				AL2005			
	ECE ↓	Brier ↓	NLL ↓	Corr(U,E) ↑	ECE ↓	Brier ↓	NLL ↓	Corr(U,E) ↑	ECE ↓	Brier ↓	NLL ↓	Corr(U,E) ↑
UST-KT	0.0970 ± 0.0071	0.1509 ± 0.0083	0.5060 ± 0.0122	0.3126 ± 0.0304	0.1122 ± 0.0083	0.2284 ± 0.0114	0.6630 ± 0.0165	0.2922 ± 0.0424	0.1516 ± 0.0102	0.2034 ± 0.0154	1.2757 ± 0.0512	0.2907 ± 0.0508
w/o Decoupling	0.1094 ± 0.0053	0.1543 ± 0.0074	0.5199 ± 0.0105	0.2508 ± 0.0405	0.1238 ± 0.0074	0.2296 ± 0.0110	0.6704 ± 0.0214	0.2606 ± 0.0315	0.1533 ± 0.0112	0.2107 ± 0.0161	1.3840 ± 0.0621	0.2814 ± 0.0526
w/o Hawkes	0.1083 ± 0.0062	0.1552 ± 0.0077	0.5298 ± 0.0110	0.2149 ± 0.0436	0.1205 ± 0.0084	0.2310 ± 0.0103	0.6807 ± 0.0212	0.2133 ± 0.0308	0.1552 ± 0.0110	0.2104 ± 0.0157	1.2655 ± 0.0600	0.1898 ± 0.0503
w/o C.Injection	0.1151 ± 0.0070	0.1621 ± 0.0089	0.5872 ± 0.0114	0.2305 ± 0.0443	0.1982 ± 0.0108	0.2411 ± 0.0106	0.8132 ± 0.0197	0.2346 ± 0.0351	0.1649 ± 0.0116	0.2236 ± 0.0177	1.2853 ± 0.0542	0.2079 ± 0.0512
w/o W.dist	0.1263 ± 0.0057	0.1711 ± 0.0068	0.5848 ± 0.0098	0.2443 ± 0.0439	0.1771 ± 0.0122	0.2777 ± 0.0149	0.7997 ± 0.0210	0.2243 ± 0.0371	0.1591 ± 0.0149	0.2284 ± 0.0143	1.4552 ± 0.0605	0.2184 ± 0.0423
w/o S.Embedding	0.1171 ± 0.0069	0.1708 ± 0.0087	0.5578 ± 0.0103	0.1947 ± 0.0429	0.1758 ± 0.0100	0.2627 ± 0.0096	0.7549 ± 0.0179	0.2046 ± 0.0342	0.1621 ± 0.0115	0.2131 ± 0.0165	1.3019 ± 0.0591	0.1898 ± 0.0458
UKT	0.1252 ± 0.0091	0.1730 ± 0.0137	0.5303 ± 0.0132	0.2443 ± 0.0552	0.1901 ± 0.0154	0.2641 ± 0.0204	0.7307 ± 0.0306	0.2404 ± 0.0491	0.1615 ± 0.0125	0.2136 ± 0.0158	1.4520 ± 0.0808	0.2242 ± 0.0510

Table 6
AUC ablation results of UST-KT on three benchmark datasets. p-values are from one-sided paired t-tests (UST-KT vs. each variant).

Model	BD2006		NIPS34		AL2005	
	AUC	p-value	AUC	p-value	AUC	p-value
UST-KT	**0.8290 ± 0.0003**	—	**0.8058 ± 0.0003**	—	**0.9337 ± 0.0013**	–
w/o Decoupling	0.8220 ± 0.0002	< 0.001***	0.8052 ± 0.0004	0.0084**	0.9313 ± 0.0011	0.0060**
w/o Hawkes	0.8195 ± 0.0005	< 0.001***	0.8020 ± 0.0004	< 0.001***	0.9307 ± 0.0015	0.0026**
w/o C.Injection	0.8193 ± 0.0006	< 0.001***	0.8023 ± 0.0005	< 0.001***	0.9304 ± 0.0014	0.0160*
w/o W.dist	0.8185 ± 0.0006	< 0.001***	0.8024 ± 0.0007	< 0.001***	0.9308 ± 0.0015	0.0251*
w/o S.Embedding	0.8187 ± 0.0009	< 0.001***	0.8035 ± 0.0008	0.0022**	0.9282 ± 0.0010	0.0017**

* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.
One-sided paired t-test, UST-KT > variant.

The calibration impact is particularly visible when covariance injection is disabled (*w/o C.Injection*): NLL increases on NIPS34 (from 0.6630 to 0.8132) and Corr(U,E) decreases accordingly, confirming that the decoupled covariance channel captures information associated with fluctuation-related unpredictability that the mean representation alone cannot recover. Disabling Hawkes-style temporal modulation (*w/o Hawkes*) produces a notable reduction in Corr(U,E) on BD2006 (from 0.3126 to 0.2149), indicating that temporally misaligned attention aggregation affects not only predictive rankings but also the alignment between predicted uncertainty and empirical errors. Replacing Wasserstein similarity (*w/o W.dist*) or stochastic embedding (*w/o S.Embedding*) further degrades calibration across datasets, reinforcing that both distributional similarity and Gaussian representations are necessary for uncertainty-informed knowledge tracing. These calibration results also corroborate the ablation findings in Table 6: modules whose removal yields only a marginal AUC drop can still produce measurable deterioration in calibration quality, which suggests that AUC saturation on mature benchmarks can mask the stabilizing role of structural components that calibration metrics remain sensitive to.

5.7. Ablation study (RQ4)

To quantify the contribution of each key design choice in UST-KT, we evaluate five ablated variants. *w/o Decoupling* removes the structural cognitive decoupling module while keeping Hawkes–Wasserstein attention. *w/o Hawkes* disables the Hawkes-inspired continuous-time modulation while retaining the decoupling module and Wasserstein-based similarity. *w/o C.Injection* disables covariance usage at the prediction layer by feeding a zero vector instead of the learned covariance, rather than removing the variance update operation inside the decoupling module. *w/o W.dist* replaces Wasserstein-distance-based similarity with standard scaled dot-product similarity in attention. *w/o S.Embedding* replaces stochastic Gaussian embeddings with deterministic item embeddings (variance set to zero, no sampling).

Table 6 reports AUC results alongside one-sided paired t-test p-values between UST-KT and each variant across five cross-validation folds. Removing any component leads to consistent performance degradation across all three datasets, and all differences reach statistical significance ($p < 0.05$ in all cases, $p < 0.001$ for the majority of comparisons on BD2006). This suggests that the improvements of UST-KT are not driven by a single module and do not arise from random variation.

The pattern of performance drops further reveals the complementary roles of each component. On BD2006, the largest drops are observed when removing distribution-aware designs, including dot-product substitution (*w/o W.dist*: 0.8185) and disabling covariance usage (*w/o C.Injection*: 0.8193), suggesting that Wasserstein similarity and covariance signals provide complementary benefits beyond mean-only representations. On AL2005, removing stochastic embedding (*w/o S.Embedding*: 0.9282) yields the most visible decrease, indicating that distributional representations are particularly beneficial for this dataset, likely because its fine-grained step-level responses make per-interaction uncertainty proxy more informative. On NIPS34, *w/o Hawkes* produces the largest drop (0.8020, $p < 0.001$), whereas *w/o Decoupling* yields the smallest decrease (0.8052, $p < 0.01$). This pattern likely reflects the dataset's highly irregular timestamps, where continuous-time modeling provides a stronger benefit than structural decoupling, and short-term behavioral noise is comparatively less dominant. This observation is also consistent with the sensitivity analysis in Fig. 3, where NIPS34 shows the strongest preference for larger γ values, confirming that smoother time-gap compression is more beneficial under highly irregular temporal patterns.

We note that a small AUC gap does not imply an unimportant component. AUC is a ranking-based metric that can saturate on mature benchmarks where strong attention baselines already capture most separable signal. In such cases, a module may primarily improve representation stability and reduce spurious oscillations rather than shift global ranking performance. Such stabilization benefits are more directly reflected in the calibration metrics of Table 5 and the smoother prediction trajectories in Fig. 5. Structural decoupling in particular is expected to show stronger AUC gains when short-term behavioral noise is more prevalent and strongly coupled with latent state updates. When noise is less dominant, decoupling primarily acts as a structural regularizer that prevents local instability without dramatically shifting global AUC. Nevertheless, its stabilizing effect remains essential for reliable uncertainty characterization, as evidenced by the consistent degradation in ECE and Corr(U,E) when it is removed.

Overall, the ablation results demonstrate that all five components each play a statistically significant and complementary role within the UST-KT framework, including structural cognitive decoupling, Hawkes-style continuous-time modulation, covariance injection, Wasserstein-based similarity, and stochastic Gaussian embedding.

5.8. Limitations and future directions

Nonetheless, our study has several limitations. The residual fluctuation extracted by the decoupling module is not guaranteed to reflect guessing or slips alone, as it may also stem from concept switching, item difficulty variation, or natural transitions across learning stages; the variance channel should therefore be understood as a useful proxy for interaction-level unpredictability rather than a fully identifiable measure of behavioral noise. In addition, the benefit of continuous-time modeling inherently depends on the quality and granularity of available timestamps.

In future work, we plan to explore more expressive uncertainty representations beyond the diagonal Gaussian approximation. Another important avenue is to develop adaptive multi-scale temporal dynamics to better capture non-monotonic forgetting patterns. Lastly, we aim to extend the framework to multi-concept and graph-structured knowledge representations to broaden the applicability of UST-KT to more realistic educational settings.

6. Conclusion

In this paper, we propose UST-KT, an uncertainty-aware probabilistic knowledge tracing framework. The proposed framework characterizes student latent knowledge states as diagonal Gaussian distributions. Through a structural cognitive decoupling module, we employ causal convolution to separate low-frequency cognitive trends from high-frequency residual fluctuations in the raw interaction sequences, and assign clear physical semantics to the Gaussian mean, representing stable mastery, and the covariance, serving as a proxy for transient behavioral uncertainty. Furthermore, a Hawkes–Wasserstein attention mechanism is introduced to dynamically modulate Wasserstein-distance-based distributional similarity using a continuous-time decay kernel, achieving deep synergy between discrete practice order and continuous-time dynamics. Extensive experiments on three benchmark datasets demonstrate that our method outperforms existing models and exhibits superior performance in multi-dimensional analyses including temporal continuity and uncertainty calibration, providing more robust user modeling and adaptive intervention support for intelligent tutoring systems.

CRediT authorship contribution statement

Xingang Zhang: Writing – review & editing, Supervision, Funding acquisition. **Xiaofan Shao:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Chuang Liu:** Validation, Investigation. **Dongyan Zhang:** Writing – original draft, Data curation, Conceptualization. **Tingzhi Li:** Investigation. **Xue Wang:** Writing – original draft, Supervision, Methodology.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT (OpenAI) to improve the readability and language of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The datasets used in this study are publicly available from the corresponding benchmark repositories. The source code of UST-KT is available at: https://github.com/USTKT/USK-KT.

References

[1] G. Abdelrahman, Q. Wang, B. Nunes, Knowledge tracing: A survey, ACM Comput. Surv. 55 (11) (2023) 1–37, http://dx.doi.org/10.1145/3569576.
[2] C.-M. Chen, H.-M. Lee, Y.-H. Chen, Personalized e-learning system using item response theory, Comput. Educ. 44 (3) (2005) 237–255, http://dx.doi.org/10.1016/j.compedu.2004.01.006.
[3] M.C. Desmarais, R.S.d. Baker, A review of recent advances in learner and skill modeling in intelligent learning environments, User Model. User-Adapt. Interact. 22 (1) (2012) 9–38, http://dx.doi.org/10.1007/s11257-011-9106-8.
[4] A.T. Corbett, J.R. Anderson, Knowledge tracing: Modeling the acquisition of procedural knowledge, User Model. User-Adapt. Interact. 4 (4) (1994) 253–278, http://dx.doi.org/10.1007/BF01099821.
[5] C. Piech, J. Bassen, J. Huang, S. Ganguli, M. Sahami, L.J. Guibas, J. Sohl-Dickstein, Deep knowledge tracing, Adv. Neural Inf. Process. Syst. 28 (2015) URL https://proceedings.neurips.cc/paper_files/paper/2015/file/bac9162b47c56fc8a4d2a519803d51b3-Paper.pdf.
[6] C.-K. Yeung, D.-Y. Yeung, Addressing two problems in deep knowledge tracing via prediction-consistent regularization, in: Proceedings of the Fifth Annual ACM Conference on Learning At Scale, 2018, pp. 1–10, http://dx.doi.org/10.1145/3231644.3231647.
[7] S. Pandey, G. Karypis, A self-attentive model for knowledge tracing, 2019, http://dx.doi.org/10.48550/arXiv.1907.06837, arXiv preprint arXiv:1907.06837.
[8] A. Ghosh, N. Heffernan, A.S. Lan, Context-aware attentive knowledge tracing, in: Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, 2020, pp. 2330–2339, http://dx.doi.org/10.1145/3394486.3403282.
[9] C. He, J. Luo, Y. Tang, G. Chen, Q. Guan, Graph neural network empowers intelligent education: A systematic review from an application perspective, IEEE Trans. Learn. Technol. 18 (2025) 1003–1020, http://dx.doi.org/10.1109/TLT.2025.3630276.
[10] S. Li, Q. Guan, L. Fang, F. Xiao, Z. He, Y. He, W. Luo, Cognitive diagnosis focusing on knowledge concepts, in: Proceedings of the 31st ACM International Conference on Information & Knowledge Management, CIKM '22, Association for Computing Machinery, New York, NY, USA, 2022, pp. 3272–3281, http://dx.doi.org/10.1145/3511808.3557096.
[11] H. Pashler, N. Cepeda, R.V. Lindsey, E. Vul, M.C. Mozer, Predicting the optimal spacing of study: A multiscale context model of memory, Adv. Neural Inf. Process. Syst. 22 (2009) URL https://proceedings.neurips.cc/paper_files/paper/2009/file/6bc24fc1ab650b25b4114e93a98f1eba-Paper.pdf.
[12] H. Ma, Y. Yang, C. Qin, X. Yu, S. Yang, X. Zhang, H. Zhu, Hd-kt: Advancing robust knowledge tracing via anomalous learning interaction detection, in: Proceedings of the ACM Web Conference 2024, 2024, pp. 4479–4488, http://dx.doi.org/10.1145/3589334.3645718.
[13] Q. Guan, X. Cheng, F. Xiao, Z. Li, C. He, L. Fang, G. Chen, Z. Gong, W. Luo, Explainable exercise recommendation with knowledge graph, Neural Netw. 183 (2025) 106954, http://dx.doi.org/10.1016/j.neunet.2024.106954.
[14] X. Cheng, Z. Zhang, J. Wang, L. Fang, C. He, Q. Guan, S. Pan, W. Luo, GraphRAG-induced dual knowledge structure graphs for personalized learning path recommendation, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 40, 2026, pp. 14610–14620, http://dx.doi.org/10.1609/aaai.v40i17.38479.
[15] W. Cheng, H. Du, C. Li, E. Ni, L. Tan, T. Xu, Y. Ni, Uncertainty-aware knowledge tracing, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 39, 2025, pp. 27905–27913, http://dx.doi.org/10.1609/aaai.v39i27.35007.
[16] H. Wu, J. Xu, J. Wang, M. Long, Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting, Adv. Neural Inf. Process. Syst. 34 (2021) 22419–22430, URL https://proceedings.neurips.cc/paper_files/paper/2021/file/bcc0d400288793e8bdcd7c19a8ac0c2b-Paper.pdf.
[17] T. Zhou, Z. Ma, Q. Wen, X. Wang, L. Sun, R. Jin, Fedformer: Frequency enhanced decomposed transformer for long-term series forecasting, in: International Conference on Machine Learning, PMLR, 2022, pp. 27268–27286, URL https://proceedings.mlr.press/v162/zhou22g.html.
[18] M. Hou, X. Li, T. Guo, Z. Liu, M. Tian, R. Luo, W. Luo, Cognitive fluctuations enhanced attention network for knowledge tracing, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 39, 2025, pp. 14265–14273, http://dx.doi.org/10.1609/aaai.v39i13.33562.
[19] J.C. Duncheon, W.G. Tierney, Changing conceptions of time: Implications for educational research and practice, Rev. Educ. Res. 83 (2) (2013) 236–272, http://dx.doi.org/10.3102/0034654313478492.
[20] Z. Liu, Q. Liu, J. Chen, S. Huang, W. Luo, simpleKT: a simple but tough-to-beat baseline for knowledge tracing, 2023, http://dx.doi.org/10.48550/arXiv.2302.06881, arXiv preprint arXiv:2302.06881.
[21] C. Wang, W. Ma, M. Zhang, C. Lv, F. Wan, H. Lin, T. Tang, Y. Liu, S. Ma, Temporal cross-effects in knowledge tracing, in: Proceedings of the 14th ACM International Conference on Web Search and Data Mining, 2021, pp. 517–525, http://dx.doi.org/10.1145/3437963.3441802.
[22] Y. Choi, Y. Lee, J. Cho, J. Baek, B. Kim, Y. Cha, D. Shin, C. Bae, J. Heo, Towards an appropriate query, key, and value computation for knowledge tracing, in: Proceedings of the Seventh ACM Conference on Learning@ Scale, 2020, pp. 341–344, http://dx.doi.org/10.1145/3386527.3405945.

[23] Y. Qian, Z. Guan, X. Wang, Z. Yang, Memory-driven attentive knowledge tracing for implicit inter-question analysis, IEEE Trans. Consum. Electron. 71 (1) (2024) 323–333, http://dx.doi.org/10.1109/TCE.2024.3512004.

[24] J. Liu, Z. Guan, X. Wang, Z. Yang, Fusing interaction transition features and cross-attention for knowledge tracing in E-learning systems, IEEE Trans. Consum. Electron. 71 (1) (2024) 761–772, http://dx.doi.org/10.1109/TCE.2024.3511994.

[25] K. Nagatani, Q. Zhang, M. Sato, Y.-Y. Chen, F. Chen, T. Ohkuma, Augmenting knowledge tracing by considering forgetting behavior, in: The World Wide Web Conference, 2019, pp. 3101–3107, http://dx.doi.org/10.1145/3308558.3313565.

[26] S. Shen, Q. Liu, E. Chen, Z. Huang, W. Huang, Y. Yin, Y. Su, S. Wang, Learning process-consistent knowledge tracing, in: Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining, 2021, pp. 1452–1460, http://dx.doi.org/10.1145/3447548.3467237.

[27] M. Chen, Q. Guan, Y. He, Z. He, L. Fang, W. Luo, Knowledge tracing model with learning and forgetting behavior, in: Proceedings of the 31st ACM International Conference on Information & Knowledge Management, CIKM '22, Association for Computing Machinery, New York, NY, USA, 2022, pp. 3863–3867, http://dx.doi.org/10.1145/3511808.3557622.

[28] B. Xu, Z. Huang, J. Liu, S. Shen, Q. Liu, E. Chen, J. Wu, S. Wang, Learning behavior-oriented knowledge tracing, in: Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2023, pp. 2789–2800, http://dx.doi.org/10.1145/3580305.3599407.

[29] K. Cheng, L. Peng, P. Wang, J. Ye, L. Sun, B. Du, DyGKT: Dynamic graph learning for knowledge tracing, in: Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2024, pp. 409–420, http://dx.doi.org/10.1145/3637528.3671773.

[30] D. Shin, Y. Shim, H. Yu, S. Lee, B. Kim, Y. Choi, Saint+: Integrating temporal features for ednet correctness prediction, in: LAK21: 11th International Learning Analytics and Knowledge Conference, 2021, pp. 490–496, http://dx.doi.org/10.1145/3448139.3448188.

[31] H. Liu, S. He, J. Li, T. Liu, R. Niu, Z. Yang, Z. Zhang, Dia4KT: Diachronic relationship learning with vigilant metricized perception for knowledge tracing, IEEE Trans. Consum. Electron. (2026) http://dx.doi.org/10.1109/TCE.2026.3681686.

[32] S. Li, S. Shen, Y. Su, X. Sun, J. Lu, Q. Mo, Z. Wu, Q. Liu, Sthkt: Spatiotemporal knowledge tracing with topological Hawkes process, Expert Syst. Appl. 259 (2025) 125248, http://dx.doi.org/10.1016/j.eswa.2024.125248.

[33] Y. Bai, X. Li, Z. Liu, Y. Huang, M. Tian, W. Luo, Rethinking and improving student learning and forgetting processes for attention based knowledge tracing models, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 39, 2025, pp. 27822–27830, http://dx.doi.org/10.1609/aaai.v39i27.34998.

[34] Q. Guan, X. Duan, K. Bian, G. Chen, J. Huang, Z. Gong, L. Fang, Kvfkt: A new horizon in knowledge tracing with attention-based embedding and forgetting curve integration, in: Proceedings of the 31st International Conference on Computational Linguistics, 2025, pp. 4399–4409, URL https://aclanthology.org/2025.coling-main.297/.

[35] J. Deng, Y. Bai, M. Hou, T. Guo, Z. Liu, W. Luo, Denoised attention and question-augmented representations for knowledge tracing, in: Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, 2025, pp. 9619–9627, http://dx.doi.org/10.24963/ijcai.2025/1069.

[36] J. Sun, S. Du, Z. Liu, F. Yu, S. Liu, X. Shen, Weighted heterogeneous graph-based three-view contrastive learning for knowledge tracing in personalized e-learning systems, IEEE Trans. Consum. Electron. 70 (1) (2023) 2838–2847, http://dx.doi.org/10.1109/TCE.2023.3293953.

[37] S. Mao, G. Li, Y. Jiang, Boosting knowledge tracing with structure-preserving adaptive contrastive learning, IEEE Trans. Consum. Electron. (2025) http://dx.doi.org/10.1109/TCE.2025.3595201.

[38] T. Guo, Y. Qin, Y. Xia, M. Hou, Z. Liu, F. Xia, W. Luo, Enhancing knowledge tracing through decoupling cognitive pattern from error-prone data, in: Proceedings of the ACM on Web Conference 2025, 2025, pp. 5108–5116, http://dx.doi.org/10.1145/3696410.3714486.

[39] X. Li, Y. Bai, T. Guo, Z. Liu, Y. Huang, X. Zhao, F. Xia, W. Luo, J. Weng, Enhancing length generalization for attention based knowledge tracing models with linear biases, in: Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24), 2024, pp. 5918–5926, http://dx.doi.org/10.24963/ijcai.2024/654.

[40] D.R. Cox, V. Isham, Point Processes, vol. 12, CRC Press, 1980.

[41] J. Stamper, Z.A. Pardos, The 2010 KDD cup competition dataset: Engaging the machine learning community in predictive learning analytics, J. Learn. Anal. 3 (2) (2016) 312–316, http://dx.doi.org/10.18608/jla.2016.32.16.

[42] Z. Wang, A. Lamb, E. Saveliev, P. Cameron, Y. Zaykov, J.M. Hernández-Lobato, R.E. Turner, R.G. Baraniuk, C. Barton, S.P. Jones, et al., Instructions and guide for diagnostic questions: The neurips 2020 education challenge, 2020, http://dx.doi.org/10.48550/arXiv.2007.12061, arXiv preprint arXiv:2007.12061.

[43] Z. Liu, Q. Liu, J. Chen, S. Huang, J. Tang, W. Luo, pyKT: a python library to benchmark deep learning based knowledge tracing models, Adv. Neural Inf. Process. Syst. 35 (2022) 18542–18555, URL https://proceedings.neurips.cc/paper_files/paper/2022/file/75ca2b23d9794f02a92449af65a57556-Paper-Datasets_and_Benchmarks.pdf.